

1. Executive Summary & Core Safety Invariant

NON-NEGOTIABLE DOMAIN INVARIANT (UK HSG47 & CDM REGULATIONS)

SafeDig re-validates upstream utility warning claims against physical drawings inside the excavation site. SAFE_MODE is permanently active. Every detected hazard is surfaced to a certified human safety engineer. Zero escaped hazards is the non-negotiable threshold. Every ambiguous, corrupted, or unreadable document must fail toward HUMAN_REVIEW or BLOCKED, NEVER toward AUTO_CLEAR. A false negative is never traded for a cleaner false-positive statistic.

2. The 5 Logical Gate Stages

Stage	Gate Range	Primary Responsibility & Safety Role
Stage 1: Pre-Flight Intake	Gates 01 - 04	Validates index row, PDF file presence, deterministic 1-to-1 mapping, and stream readability.
Stage 2: Knowledge Resolution	Gates 05 - 07	Resolves utility undertaker (UKPN, SGN, Cadent, etc.), warning catalogue, and color/stroke legend.
Stage 3: Spatial AOI Boundary	Gates 08 - 09	Detects red dig polygon (Area of Interest), validates closed Shapely geometry & coordinate bounds.
Stage 4: CV & Asset Reconcile	Gates 10 - 13	Dual-engine vector & raster CV scan, spatial intersection math, zero missed critical warnings.
Stage 5: QA, Evidence & Audit	Gates 14 - 17	Raster DPI check, provider safety buffers (HSG47), high-res evidence bundle, SQLite audit commit.

3. Summary of all 17 Release Gates

Gate ID	Formal Gate Name	Trigger Condition	Failure Route
Gate 01	01_INDEX_VALID	Index record exists and has vali...	HUMAN_REVIEW
Gate 02	02_MAP_EXISTS	Physical PDF exists on filesyste...	BLOCKED
Gate 03	03_MAPPING_VALID	Deterministic 1-to-1 mapping (UN...	BLOCKED
Gate 04	04_MAP_READABLE	PDF format readable (VECTOR, RAS...	HUMAN_REVIEW
Gate 05	05_PROVIDER_RESOLVED	Statutory utility undertaker suc...	HUMAN_REVIEW
Gate 06	06_CATALOGUE_RESOLVED	Provider master warning catalogu...	HUMAN_REVIEW
Gate 07	07_LEGEND_RESOLVED	Color, line stroke, and symbol l...	HUMAN_REVIEW
Gate 08	08_AOI_RESOLVED	Excavation boundary (Area of Int...	HUMAN_REVIEW
Gate 09	09_AOI_VALIDATION_COMPLETED	AOI polygon valid Shapely geomet...	HUMAN_REVIEW
Gate 10	10_INDEPENDENT_SCAN_COMPLETED	Dual vector and raster CV scan e...	HUMAN_REVIEW
Gate 11	11_RECONCILIATION_COMPLETED	Spatial intersection math comput...	HUMAN_REVIEW
Gate 12	12_NO_UNRESOLVED_CRITICAL_WARNING	Zero missed warnings & no unrevi...	HUMAN_REVIEW
Gate 13	13_NO_DETECTOR_DISAGREEMENT	Upstream claim and CV detection ...	HUMAN_REVIEW
Gate 14	14_NO_IMAGE_QUALITY_ISSUE	Resolution >= 200 DPI, contrast ...	HUMAN_REVIEW
Gate 15	15_PROVIDER_RULES_PASS	Provider safety standoff buffers...	HUMAN_REVIEW
Gate 16	16_EVIDENCE_COMPLETE	AOI crop, full sheet, overlay, G...	HUMAN_REVIEW
Gate 17	17_AUDIT_PERSISTED	SHA-256 hashes, version, timesta...	BLOCKED

Stage 1: Document Intake & Pre-Flight Validation

Gate: 01_INDEX_VALID · Index Record Verification Code: <code>index_record is not None and bool(index_record.utility_name)</code> Parses index.xlsx or enquiry summary. Ensures the utility enquiry line is non-empty, contains an official utility name, and maps to an active search job. Prevents orphaned drawings. Consequence: Fails to HUMAN_REVIEW. Job is flagged as unverified administrative entry.
Gate: 02_MAP_EXISTS · Physical Drawing Presence on Filesystem Code: <code>document is not None and not document.is_corrupted</code> Verifies that the target PDF exists on the disk, has file size > 0 bytes, contains valid %PDF- header magic, and can be opened by PyMuPDF. Solves portable path resolution. Consequence: CRITICAL: Immediate early-exit to BLOCKED. If utility stated assets are present but map file is missing, excavation is halted.
Gate: 03_MAPPING_VALID · Deterministic 1-to-1 Document Resolution Code: <code>index_record.resolution_status in [UNIQUE, EXCLUDED]</code> Ensures that exactly one drawing corresponds to the enquiry record. Excludes customer letters or safety brochures. Detects AMBIGUOUS situations where multiple drawings compete. Consequence: If status is AMBIGUOUS, early-exits to BLOCKED. Never allows the system to guess which map to inspect.
Gate: 04_MAP_READABLE · PDF Stream & Modality Usability Code: <code>document is not None and document.modality != PDFModality.UNREADABLE</code> Analyzes page count, encryption keys, and internal stream objects. Classifies drawing as VECTOR, RASTER, or HYBRID. Rejects corrupted binary streams or unreadable scans. Consequence: Fails to HUMAN_REVIEW / BLOCKED. Flagged for manual document re-acquisition.

Stage 2: Provider & Knowledge Resolution

Gate: 05_PROVIDER_RESOLVED · Statutory Undertaker Identification Code: <code>bool(index_record.utility_name.strip())</code> Resolves provider to registered UK statutory body (UKPN, SGN, Cadent, NGET, NGT, Virgin Media, Openreach, Thames Water, GTC). Normalizes regional subsidiaries into canonical IDs. Consequence: Fails to HUMAN_REVIEW. Cannot clear without identifying the statutory asset owner.
Gate: 06_CATALOGUE_RESOLVED · Master Warning Catalogue Activation Code: <code>warning_catalogue.is_active and provider in catalogue</code> Loads provider-specific hazard definitions, statutory voltages (11kV/33kV/132kV), gas pressure categories (LP, MP, IP, HP), and required safety action codes into active memory. Consequence: Fails to HUMAN_REVIEW. Ensures policy operates with full regulatory taxonomy.
Gate: 07_LEGEND_RESOLVED · Cartographic Legend Profile Resolution Code: <code>legend_profile is not None</code> Loads RGB color tolerance bands (Delta E <= 28), stroke line patterns (dashed, dotted, solid), and symbol libraries (substations, link boxes, valves, chambers) for this utility provider. Consequence: Fails to HUMAN_REVIEW. Without resolved legend profile, automated CV is suspended.

Stage 3: Spatial AOI & Excavation Boundary

Gate: 08_AOI_RESOLVED · Excavation Site Boundary Extraction

Code: `aoi is not None and aoi.is_valid`

Runs spatial AOI detector to extract red boundary lines, red diagonal hatching, corner pins, or site plan bounding boxes. Maps drawing coordinate space to physical dig site envelope.

Consequence: Fails to HUMAN_REVIEW. Safe clearance impossible if the dig perimeter is undetermined.

Gate: 09_AOI_VALIDATION_COMPLETED · Topological & Geometric Validation (Shapely)

Code: `aoi is not None and len(aoi.coordinates) >= 4`

Verifies coordinate closure (`coords[0] == coords[-1]`), vertex count (`>= 4`), valid non-degenerate surface area, and checks that polygon coordinates reside within sheet canvas bounds.

Consequence: Fails to HUMAN_REVIEW. Prevents distorted, infinite, or zero-area polygons.

Stage 4: Computer Vision & Asset Reconciliation

Gate: 10_INDEPENDENT_SCAN_COMPLETED · Dual-Engine Vector & Raster Spatial Scan

Code: `vector_scan_completed and raster_scan_completed`

Executes PyMuPDF vector path geometric extraction combined with OpenCV 300 DPI raster color filtering and contour analysis. Aggregates all detected buried assets into a unified catalog.

Consequence: Fails to HUMAN_REVIEW / BLOCKED. Guardrail against scanner execution crashes.

Gate: 11_RECONCILIATION_COMPLETED · Spatial Intersection & Upstream Reconciliation

Code: `reconciliation is not None`

Computes exact 2D geometric intersection between AOI polygon (plus 5m/15m safety buffer) and detected utility vectors. Cross-references detected assets against upstream enquiry claims.

Consequence: Fails to HUMAN_REVIEW. Documents reconciliation outcome (Match, Missed Warning, Clean).

Gate: 12_NO_UNRESOLVED_CRITICAL_WARNING · Critical Life-Safety Asset Clearance Gate

Code: `not (outcome == MISSED_WARNING or (outcome == MATCH and severity in [CRITICAL, HIGH]))`

CORE SAFETY SHIELD: Fails if an omitted hazard is detected in the AOI (MISSED_WARNING), or if a confirmed High Voltage cable (11kV+) or High Pressure Gas main is present.

Consequence: MANDATORY HUMAN ESCALATION. Cannot be auto-cleared. Human safety engineer sign-off required.

Gate: 13_NO_DETECTOR_DISAGREEMENT · Consensus & Contradiction Detection

Code: `reconciliation.outcome in [CONFIRMED_CLEAN, MATCH]`

Flags contradictory evidence between upstream claim ('No assets') and physical drawing ('Vector present'), or false positives where upstream claimed assets but drawing shows clear.

Consequence: Fails to HUMAN_REVIEW with highlighted difference overlay.

Stage 5: Quality Assurance, Evidence & Immutable Audit

Gate: 14_NO_IMAGE_QUALITY_ISSUE · Raster Image Quality & DPI Verification

Code: [document is not None and not document.is_corrupted](#)

Inspects map rasterization quality (>= 200 DPI). Verifies that compression artifacts, low contrast, or scanning distortion do not obscure hairline utility vectors.

Consequence: Fails to HUMAN_REVIEW. Ensures visual clarity for both algorithms and engineers.

Gate: 15_PROVIDER_RULES_PASS · Statutory Safety Standoff Buffer Evaluation

Code: [provider_rules_evaluated and buffer_compliance_verified](#)

Enforces UK HSG47 500mm hand-dig safety boundary, 3m electrical safety standoff, and Cadent/SGN 15m high-pressure gas clearance zones.

Consequence: Fails to HUMAN_REVIEW. Prevents mechanical excavation inside statutory clearance envelopes.

Gate: 16_EVIDENCE_COMPLETE · Forensic Evidence Package Verification

Code: [evidence_pkg.is_complete](#)

Validates that the complete evidence dossier exists on disk: high-res AOI crop, whole-sheet context plan, neon vector overlay rendering, and GeoJSON geometry payload.

Consequence: Fails to HUMAN_REVIEW if any evidence artifact is missing or zero bytes.

Gate: 17_AUDIT_PERSISTED · Cryptographic Audit Persistence (SQLite & SHA-256)

Code: [audit_record_persisted and sha256_fingerprint_verified](#)

Computes SHA-256 hashes of original PDF, index record, detection GeoJSON, and timestamps. Commits immutable audit record to safedig.db for HSE legal defensibility.

Consequence: CRITICAL: Clearance blocked if audit snapshot cannot be locked in the database.

Policy Engine Decision Ladder (src/policy/engine.py)

Ladder Step	Evaluation Criteria	Final Decision
Step 1: Early Exit Clean	If index_record.is_asset_present == False (Status='No')	AUTO_CLEAR
Step 2: Early Exit Blocked	If document is None or document.is_corrupted	BLOCKED
Step 3: Early Exit Ambiguous	If index_record.resolution_status == AMBIGUOUS	BLOCKED
Step 4: Missed Warning Check	If reconciliation.outcome == MISSED_WARNING	HUMAN_REVIEW
Step 5: False Positive Check	If reconciliation.outcome == POSSIBLE_FALSE_POSITIVE	HUMAN_REVIEW
Step 6: Critical Hazard Check	If outcome == MATCH and Gate 12 Fails (HV / HP Gas)	HUMAN_REVIEW
Step 7: Geometry/Legend Check	If Gate 07 (Legend) or Gate 08 (AOI) Failed	HUMAN_REVIEW
Step 8: Final Gate Sweep	If ALL 17 Gates passed and evidence is complete	AUTO_CLEAR

Operational Case Studies & Real-World Gate Behavior

Scenario A: Clean Telecom Conduit Outside AOI Enquiry: Openreach / BT copper/fiber conduits. CV Scan: Detected lines 14.2 meters north of the excavation polygon. Gate Behavior: Gate 01-17 all evaluate to True. Gate 11 computes zero intersection. Gate 12 passes. Outcome: System issues AUTO_CLEAR. Safe for automated excavation release.	AUTO_CLEAR
Scenario B: Undisclosed 11kV High-Voltage Cable Inside AOI Enquiry: Upstream declaration stated 'No affected assets in enquiry boundary'. CV Scan: PyMuPDF vector engine extracts red dashed vector (RGB #FF0000) intersecting site center. Gate Behavior: Gate 12 FAILS (MISSED_WARNING). Gate 13 FAILS (Disagreement between upstream & CV). Outcome: IMMEDIATE HUMAN_REVIEW. SafeDig prevents an excavation team from striking an active 11kV feeder.	HUMAN_REVIEW
Scenario C: Degraded Drawing with Ambiguous Candidate Maps Enquiry: SGN Gas distribution with two competing PDFs in folder. Inventory Matcher: Both PDFs share identical high confidence scores (DocumentResolutionStatus.AMBIGUOUS). Gate Behavior: Gate 03 FAILS. PolicyEngine triggers early-exit blocking condition. Outcome: BLOCKED. System halts clearance until a human operator specifies the authoritative drawing.	BLOCKED

Audit Traceability & QA Workspace Integration

- Interactive QA Workspace (src/qa/workspace.py & src/api/static/):
- Safety engineers inspect any flagged job via the web UI at /#qa-workspace.
 - The workspace visualizes all 17 gates dynamically in a status grid with green checkmarks or red failure badges.
 - Clicking any gate filters the vector overlay to highlight the exact visual evidence or geometry violation.
 - Human review overrides (Accept Risk / Reject Permit) require safety engineer credentials and are permanently logged into safedig.db alongside the original gate evaluation snapshot.
- Cryptographic Defense:
- Every release decision produces a tamper-proof SHA-256 digital signature recorded in safedig.db.
 - In the event of an HSE statutory inquiry, the complete execution trajectory (PDF hash, AOI coordinates, vector intersection coordinates, and all 17 gate boolean states) can be reproduced bit-for-bit.